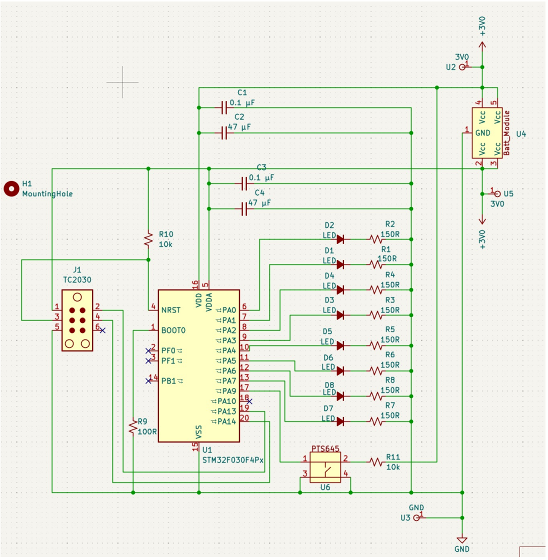
V₂



Electrical characteristics **STM32F030x4/x6/x8/xC**

Table 19. Current characteristics

Symbol	Ratings	Max.	Unit
ΣI_{VDD}	Total current into sum of all VDD power lines (source)(1)	120	mA
ΣI_{VSS}	Total current out of sum of all VSS ground lines (sink)(1)	-120	
$I_{VDD(PIN)}$	Maximum current into each VDD power pin (source)(1)	100	
$I_{VSS(PIN)}$	Maximum current out of each VSS ground pin (sink)(1)	-100	
$I_{IO(PIN)}$	Output current sunk by any I/O and control pin	25	
	Output current source by any I/O and control pin	-25	
$\Sigma I_{IO(PIN)}$	Total output current sunk by sum of all I/Os and control pins(2)	80	
	Total output current sourced by sum of all I/Os and control pins(2)	-80	
$I_{INJ(PIN)}$(3)	Injected current on FT and FTf pins	-5/+0(4)	
	Injected current on TC and RST pin	± 5	
	Injected current on TTa pins(5)	± 5	
$\Sigma I_{INJ(PIN)}$	Total injected current (sum of all I/O and control pins)(6)	± 25	

80 mA maximum draw

↳ 70% absolute max = 56 mA

↳ $\frac{80\text{ mA}}{56\text{ mA}} = 1.429\times$ SF

$\frac{56\text{ mA}}{8\text{ LEDs}} = 7\text{ mA/LED}$

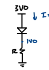
↳ 43 RLI, ≈ 27 mcd

↳ 2.0 V FV

4. ABSOLUTE MAXIMUM RATINGS *2

Power Dissipation	PD (Ta=25℃)	· · · · · · · · · ·	54mW
Forward Current	IF (Ta=25℃)	· · · · · · · · · ·	20mA
Peak Forward Current	IFP (Ta=25℃)	· · · · · · · · · ·	100mA 1)
Reverse Voltage	VR (Ta=25℃)	· · · · · · · · · ·	5V
Operation Temperature	Topr	· · · · · · · · · ·	-40～+85℃
Storage Temperature	Tstg	· · · · · · · · · ·	-40～+100℃

1) Duty 1/10, 1kHz



$I_R = \frac{(1.0-0)V}{R} = .007A$

$R = 142.9\,\Omega \rightarrow 150\,\Omega$

$P_{LED} = IV$

$= (.007\text{ A})(2.0\text{ V})$

$= .014\text{ W}$

$= 14\text{ mW}$

$P_{Res} = I^2R$

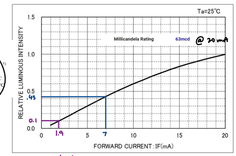
$= (.007\text{ A})^2(150\,\Omega)$

$= .00735\text{ W}$

Resistance	150 Ohms
Tolerance	±5%
Power (Watts)	0.1W, 1/10W

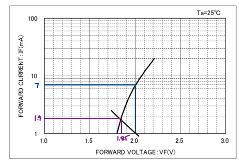
↳ $\frac{.1\text{ W}}{.00735\text{ W}} = 13.605\times$ SF

RELATIVE LUMINOUS INTENSITY - FORWARD CURRENT



(0.1)(63 mcd) = 6.3 mcd

FORWARD CURRENT - FORWARD VOLTAGE



$1000 = \frac{3-V_f}{I}$

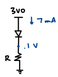
$I = \frac{3-V_f}{1000}$

↳ (1.85 V, 1.8 mA)

* chose R = 3000 Ω

Forward current ~.2 mA

APT1608LVBC/D



$.007\text{ A} = \frac{.1\text{V} - 0\text{V}}{R}$

$R = \frac{1\text{ V}}{.007\text{ A}} = 143\,\Omega$

$I_R = \frac{.1\text{V} - 0\text{V}}{15\,\Omega} = 6.7\text{ mA}$

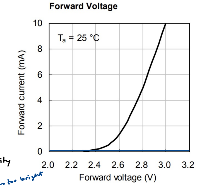
↳ ~3.5× luminous intensity

3.5(24 mcd) = 84 mcd

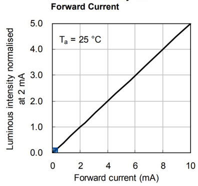
V_1 → too bright

Millicandela Rating 24mcd

Forward Current vs. Forward Voltage



Luminous Intensity vs. Forward Current



$P_{LED} = IV$

$= (.007\text{ A})(2.9\text{ V})$

$= .0203\text{ W}$

$= 20.03\text{ mW}$

ABSOLUTE MAXIMUM RATINGS at T_A=25°C

Parameter	Symbol	Value	Unit
Power Dissipation	P_D	120	mW
Reverse Voltage	V_R	5	V
Junction Temperature	T_j	115	°C
Operating Temperature	T_{op}	-40 to +85	°C
Storage Temperature	T_{stg}	-40 to +85	°C
DC Forward Current	I_F	30	mA
Peak Forward Current	I_{FM} [1]	100	mA
Electrostatic Discharge Threshold (HBM)	-	250	V
Thermal Resistance (Junction / Ambient)	$R_{th\ JA}$ [2]	510	°C/W
Thermal Resistance (Junction / Solder point)	$R_{th\ JS}$ [2]	350	°C/W

15 Ω

$I_R = \frac{.1\text{V} - 0\text{V}}{15\,\Omega} = 6.7\text{ mA}$

8(6.7 mA) = 53.6 mA

$\frac{53.6\text{ mA}}{80\text{ mA}} = 67\%$ AMR

$P_R = \frac{V^2}{R}$

$= \frac{(.1V)^2}{15}$

$= .67\text{ mW}$

Filtering Capacitor:

$Z = \frac{1}{2\pi f c}$

$f_1 = 1\text{ MHz}$

$Z_1 = \frac{1}{2\pi(1e6\text{ Hz})(1e\text{-}7\text{ F})} = 1.59\,\Omega$

$f_2 = 10\text{ MHz}$

$Z_2 = \frac{1}{2\pi(1e7\text{ Hz})(1e\text{-}7\text{ F})} = 159\,\Omega$

* lower impedance = better flow to GND

Bulk Capacitor:

Load current = 30% Abs. Max current

= 30 mA

$I_{batt} = 20\text{ mA}$

Electrical characteristics

- Nominal capacity (15k Ohm Cont., 2.0 V cut-off) 245 mAh
- Open circuit voltage (at + 20 °C) 3.0 V
- Standard Continuous Discharge Current 0.5 mA
- Maximum Continuous Discharge Current 6 mA
- Maximum Pulse Discharge Current at 1 sec 20 mA
- Nominal Energy 735 mWh
- AC Impedance @ 1kHz 20 Ohm

$I_{Load} = I_{batt} + I_{cap}$

$I_{cap} = I_{Load} - I_{batt}$

$= 30\text{ mA} - 20\text{ mA}$

$= 10\text{ mA}$

$C = \frac{Q}{V}$

$\Delta V = \frac{Q}{C} = \frac{It}{C}\Big|_{C=47\mu F}$

$= \frac{(.01\text{ A})(.002\text{ s})}{47e\text{-}6\text{ F}}$

$= .425$

$V = 3.0 - .425 = 2.575\text{ V} > 2.4\text{ V}$

$V = 3.0 + .425 = 3.425\text{ V} < 3.6\text{ V}$

Table 33. HSE oscillator characteristics

Symbol	Parameter	Conditions(1)	Min(2)	Typ	Max(2)	Unit
I_{DD}	HSE current consumption	During startup(3)	-	-	8.5	mA
		V_{DD} = 3.3 V, Rm = 45 Ω, CL = 10 pF@8 MHz	-	0.5	-	
		V_{DD} = 3.3 V, Rm = 30 Ω, CL = 20 pF@32 MHz	-	1.5	-	
g_m	Oscillator transconductance	Startup	10	-	-	mA/V
$t_{SU(HSE)}$(4)	Startup time	V_{DD} is stabilized	-	2	-	ms

Table 2. STM32F030x4/x6/x8/xC family device features and peripheral counts

Peripheral		STM32 F030F4	STM32 F030K6	STM32 F030C6	STM32 F030C8	STM32 F030CC	STM32 F030R8	STM32 F030RC
Flash (Kbytes)		16	32	32	64	256	64	256
SRAM (Kbytes)		4			8	32	8	32
Timers	Advanced control	1 (16-bit)						
	General purpose	4 (16-bit)(1)			5 (16-bit)			
	Basic	-			1 (16-bit)(2)	2 (16-bit)	1 (16-bit)(2)	2 (16-bit)
Comm. interfaces	SPI	1(3)			2			
	I^2C	1(4)			2			
	USART	1(5)			2(6)	6	2(6)	6
12-bit ADC (number of channels)		1 (9 ext. +2 int.)	1 (10 ext. +2 int.)	1 (10 ext. +2 int.)	1 (10 ext. +2 int.)	1 (10 ext. +2 int.)	1 (16 ext. +2 int.)	1 (16 ext. +2 int.)
GPIOs		15	26	39	39	37	55	51
Max. CPU frequency		48 MHz						
Operating voltage		2.4 to 3.6 V						
Operating temperature		Ambient operating temperature: -40°C to 85°C Junction temperature: -40°C to 105°C						
Packages		TSSOP20	LQFP32	LQFP48			LQFP64	